

# Short Answer Test — Model Answers (Q1–Q5)

Neural Networks & Machine Learning | Prepared per Rubric Criteria (Conceptual Understanding, Accuracy, Application, Completeness, Clarity)

## Question 1

(5 Marks)

A simple perceptron classifies pass/fail using study hours and attendance (weights 0.6, 0.4; bias -0.5). For study hours = 2, attendance = 1, find the net input, apply the step function, and conclude.

### Concept

A perceptron computes a weighted sum of its inputs plus a bias (the net input), then passes this value through a step activation function that outputs 1 if the net input is greater than or equal to 0, and 0 otherwise.

### Calculation

```
net = (w1 × x1) + (w2 × x2) + bias
net = (0.6 × 2) + (0.4 × 1) + (-0.5)
net = 1.2 + 0.4 - 0.5
net = 1.1
```

### Applying the Step Activation Function

```
f(net) = 1, if net ≥ 0
f(net) = 0, if net < 0
Since net = 1.1 ≥ 0 → output = 1
```

**Final Answer: Net input = 1.1; Output = 1 → The student is predicted to PASS.**

## Question 2

(5 Marks)

A single neuron trained with the delta rule has initial weight 0.5, learning rate 0.1, input = 2, actual output = 0.6, target output = 1. Find the error, weight update, and new weight.

### Concept

The delta rule adjusts a weight in proportion to the error between the target and actual output, scaled by the learning rate and the input value. It is the basis of gradient-descent-style weight correction used in backpropagation.

### Calculation

```
Error (δ) = target - actual = 1 - 0.6 = 0.4

Δw = η × δ × x
Δw = 0.1 × 0.4 × 2
Δw = 0.08

New weight = old weight + Δw = 0.5 + 0.08 = 0.58
```

**Final Answer: Error = 0.4; Weight update Δw = 0.08; New weight = 0.58**

## Question 3

(5 Marks)

A disease-prediction model gives TP=180, TN=150, FP=20, FN=50. Compute accuracy, precision, sensitivity (recall), and specificity, and discuss reliability for medical diagnosis.

### Concept

A confusion matrix summarises classifier performance in terms of correctly and incorrectly predicted positive/negative cases. Different metrics highlight different types of error — this matters greatly in medical diagnosis, where missing a true case (false negative) can be far costlier than a false alarm (false positive).

Calculation

```
Total = TP+TN+FP+FN = 180+150+20+50 = 400

Accuracy = (TP+TN)/Total = (180+150)/400 = 330/400 = 0.825 = 82.5%

Precision = TP/(TP+FP) = 180/200 = 0.90 = 90%

Sensitivity (Recall) = TP/(TP+FN) = 180/230 = 0.7826 ≈ 78.26%

Specificity = TN/(TN+FP) = 150/170 = 0.8824 ≈ 88.24%
```

Discussion — Reliability for Medical Diagnosis

Overall accuracy (82.5%) and precision (90%) look strong, but sensitivity/recall is comparatively lower (≈78.3%), meaning the model misses about 22% of actual disease-positive patients (50 false negatives). In medical diagnosis, false negatives are especially dangerous because a sick patient may go undiagnosed and untreated. Therefore, despite a respectable overall accuracy, this model would need improvement in recall/sensitivity — e.g., via threshold tuning, class re-balancing, or cost-sensitive training — before it can be considered fully reliable for clinical use.

**Final Answer: Accuracy = 82.5%, Precision = 90%, Recall (Sensitivity) ≈ 78.26%, Specificity ≈ 88.24%. Reliable overall, but the relatively lower recall means it risks missing real disease cases — a critical concern in medical diagnosis.**

Question 4 (5 Marks)

A Genetic Algorithm maximizes  $f(x) = x^2$  with initial population {2, 4, 6, 8}. Compute fitness, select the best individuals, perform crossover by exchanging the least significant bits, and explain the role of mutation.

Step 1: Fitness Evaluation

| Individual (x) | Binary (4-bit) | Fitness $f(x)=x^2$ |
|----------------|----------------|--------------------|
| 2              | 0010           | 4                  |
| 4              | 0100           | 16                 |
| 6              | 0110           | 36                 |
| 8              | 1000           | 64                 |

Step 2: Selection

Since the goal is maximization, individuals with the highest fitness are most likely to be selected as parents. The two fittest individuals are 8 (fitness 64) and 6 (fitness 36) — these are chosen as the parent chromosomes for reproduction.

Step 3: Crossover (Exchanging Least Significant Bits)

```
Parent 1: 8 = 1000
Parent 2: 6 = 0110

Exchange the least significant bit (LSB) of each parent:
Offspring 1 = 1000 (LSB from Parent 2) → 1000 = 8
Offspring 2 = 0110 (LSB from Parent 1) → 0110 = 6
```

In this particular pair the LSBs happen to coincide (both 0), so the offspring bit-strings equal the parents in this instance. If the least significant two bits were exchanged instead (e.g., 1000  $\leftrightarrow$  0110  $\rightarrow$  offspring 1010 = 10 and 0100 = 4), new, potentially fitter individuals such as  $x = 10$  (fitness 100) could emerge — illustrating how crossover point selection affects offspring diversity.

#### Step 4: Role of Mutation

Mutation randomly flips one or more bits in a chromosome (e.g., 1000  $\rightarrow$  1001) with a small probability. Its role is to:

- Introduce new genetic material that crossover alone cannot generate, preventing the population from getting stuck at a local optimum.
- Maintain diversity in the population across generations, which improves the algorithm's ability to explore the full search space.
- Provide a mechanism to recover genetic information that may have been lost during selection or crossover.

**Final Answer: Fitness values = 4, 16, 36, 64 for  $x = 2, 4, 6, 8$  respectively. Best candidates selected:  $x = 8$  and  $x = 6$ . Crossover on LSBs produces offspring 1000 (8) and 0110 (6) in this case; mutation is essential for maintaining diversity and avoiding premature convergence.**

## Question 5

(5 Marks)

*Training accuracy rises steadily from 60% to 95%; validation accuracy rises from 58% to 72% and then falls to 68%. Analyse the trend, identify the issue, explain why it occurs, suggest two techniques to fix it, and discuss the significance of validation accuracy.*

### Trend Analysis

Training accuracy improves continuously across all epochs, while validation accuracy improves only up to a point and then declines even as training accuracy keeps rising. This diverging pattern — training performance improving while validation performance worsens — is the classic signature of **overfitting**.

### Why Overfitting Occurs

As training continues beyond the optimal point, the model — especially a deep network with many parameters — begins to memorise the specific patterns and noise in the training data rather than learning generalisable features. The network's high capacity allows it to fit training examples almost perfectly, but these memorised, overly-specific patterns do not transfer well to unseen validation data, so validation accuracy stagnates and then drops.

### Two Techniques to Address Overfitting

- **Early Stopping** — Monitor validation accuracy/loss during training and stop (or restore the best checkpoint) as soon as validation performance stops improving, before the model starts overfitting the training set.
- **Regularization / Dropout** — Apply L1/L2 weight regularization to penalise overly large weights, or use dropout to randomly deactivate neurons during training, forcing the network to learn more robust, distributed representations instead of relying on specific memorised paths.

(Other valid techniques include data augmentation and reducing model complexity.)

### Significance of Validation Accuracy

Validation accuracy is computed on data the model has not directly trained on, so it estimates how well the model is likely to generalise to new, real-world data. It is essential for detecting overfitting, guiding hyperparameter tuning (e.g., learning rate, number of layers, regularization strength), and deciding when to stop training — making it a far more trustworthy indicator of true model quality than training accuracy alone.

**Final Answer: The issue is OVERFITTING — the model memorises training data after a certain epoch, hurting generalisation. Fix with early stopping and regularization/dropout. Validation accuracy is critical because it reflects real-world generalisation, not just memorisation.**

Note: Each answer above addresses all five rubric criteria — Conceptual Understanding, Accuracy of Answer, Application of Knowledge, Completeness of Response, and Clarity & Presentation — to align with full (Level 4 / Exemplary) marks.